



Smart Email & Letter Generator

Project for a Generative AI course

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The Challenge of Professional Writing

Professional writing is a critical skill in jobs, business, and education — yet it remains a significant barrier for many people. Whether drafting a formal email, composing a cover letter, or writing a complaint, the demands of tone, grammar, and structure can be overwhelming.

Lack of Writing Skills

Many users struggle to structure professional documents effectively, leading to poor first impressions.

Grammar Mistakes

Errors in grammar and spelling undermine credibility and professionalism in written communication.

Wrong Tone

Striking the right tone — formal, friendly, assertive — is difficult without experience or guidance.

Time Consumption

Crafting a well-written document from scratch can take hours, reducing productivity significantly.

This project was built to solve all of these problems using the power of AI — making professional writing accessible to everyone.

Problem Statement

In today's professional world, individuals are regularly required to produce a wide variety of written documents. Manual writing demands considerable time, effort, and expertise — resources that many students and professionals simply do not have in abundance.

Documents People Need to Write

- Professional emails to colleagues, clients, and managers
- Cover letters for job applications
- Formal job application letters
- Complaint letters to organisations or service providers
- Thank-you letters following interviews or meetings

The Core Problem

Each document type has its own conventions, structure, and expected tone. Without proper guidance or writing experience, producing these documents correctly is a genuine challenge.

Manual writing also introduces inconsistency — the same person may write very differently depending on their mood, time pressure, or familiarity with the subject. This leads to unprofessional output that can damage opportunities.



Time and skill barriers prevent many people from communicating professionally and effectively.

Our Proposed Solution

We developed an AI-powered web application that removes the barriers of professional writing entirely. By combining intelligent language models with a simple, intuitive interface, the tool generates polished, ready-to-use documents in seconds.



Instant Generation

Documents are generated instantly — no waiting, no manual drafting. Users receive a complete, professional output within moments of submitting their details.



Resume-Aware Content

The application uses the user's uploaded resume to personalise the generated content, ensuring relevance and accuracy in every document.



Professional Quality

Every output meets professional writing standards — correct grammar, appropriate tone, and proper structure — every single time.



Ready-to-Use Output

The generated document is immediately ready to send or submit, saving users significant time and effort in their professional endeavours.

Technologies Used

The Smart Email & Letter Generator is built on a carefully selected stack of modern, powerful technologies that work together to deliver a seamless AI-driven experience.

1

Python

The core programming language powering the entire application backend and logic.

2

Streamlit

A rapid web application framework for Python, enabling a clean and interactive user interface without complex frontend development.

3

Groq API + Llama 3.1 8B Instant

The AI engine behind document generation — Groq's ultra-fast inference paired with Meta's Llama 3.1 8B model for high-quality language output.

4

PyPDF2 & python-docx

Libraries for parsing uploaded PDF resumes and generating downloadable Word documents from the AI output.

5

Google Colab & Cloudflare Tunnel

Google Colab provides the cloud compute environment, while Cloudflare Tunnel exposes the application publicly without requiring a dedicated server.

Key Features

The application has been designed with both functionality and user experience at its core. Every feature serves a specific purpose in making professional writing faster, easier, and more effective.



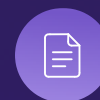
Professional UI

A clean, intuitive interface that guides users through the document creation process with minimal friction.



Resume Upload

Users can upload their CV or résumé in PDF format, which the AI uses to personalise and contextualise the generated content.



Multiple Document Types

Supports emails, cover letters, job applications, complaint letters, thank-you letters, and more — all from a single platform.



Tone & Length Selection

Users can select the desired tone (formal, friendly, assertive) and output length to match the specific context of their document.

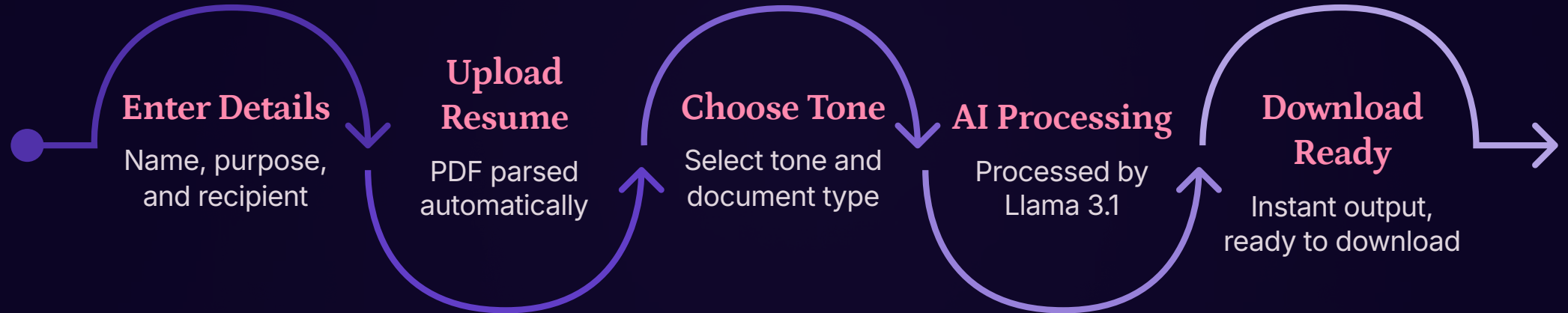


Download Output

Generated documents can be downloaded directly, ready to send or submit without any further editing required.

How It Works

The working process is designed to be straightforward and efficient. From entering basic details to receiving a polished document, the entire journey takes only a matter of seconds.



The AI processes all inputs — user details, resume content, selected tone, and document type — into a structured prompt that is sent to the Groq-hosted Llama 3.1 8B model. The model returns a fully formed, professional document in real time, which is then displayed and made available for download. The entire pipeline is optimised for speed and accuracy, ensuring a seamless experience from start to finish.

Advantages

The Smart Email & Letter Generator offers a compelling set of advantages over traditional manual writing, making it a valuable tool for a wide range of users — from students applying for their first job to seasoned professionals managing high-volume correspondence.

Who Benefits Most

The tool is particularly valuable for:

- Students applying for internships and graduate roles
- Job seekers crafting cover letters and applications
- Professionals managing formal correspondence
- Non-native English speakers needing polished output
- Anyone short on time but needing quality writing

Core Advantages



Fast Writing

Documents that would take 30–60 minutes to write manually are generated in seconds.



Professional Quality

Consistent grammar, structure, and tone in every output — no proofreading required.



Personalised Content

Resume-aware generation ensures each document is tailored to the individual user.

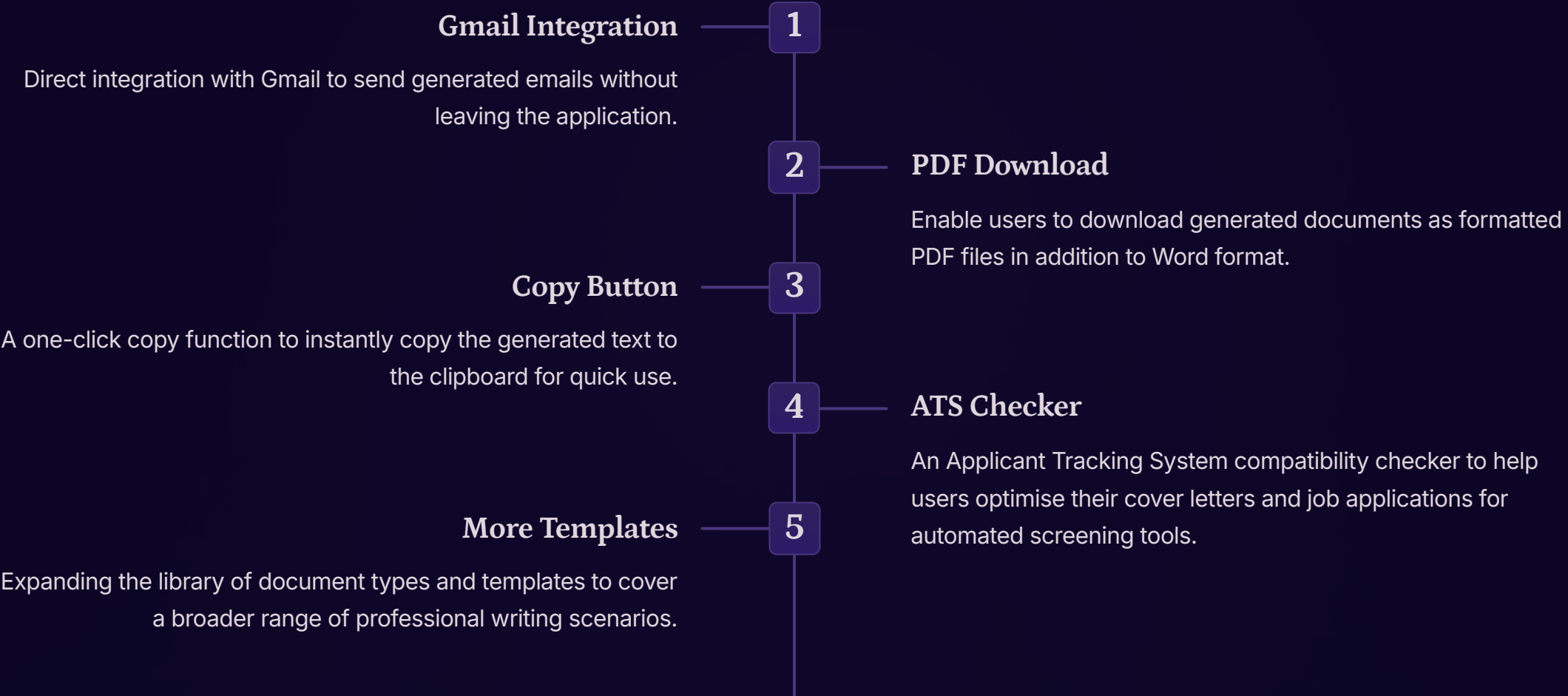


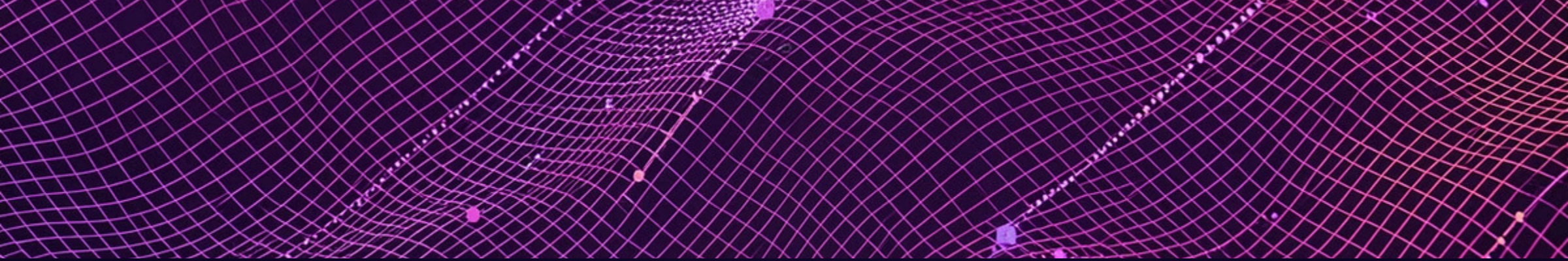
Easy to Use

No technical knowledge required — the interface is simple and accessible to all.

Future Scope

The current version of the Smart Email & Letter Generator is a strong foundation, but there is significant potential to expand its capabilities and reach. The following enhancements are planned for future development phases.





Conclusion

The **Smart Email & Letter Generator** is a practical, AI-powered tool that fundamentally improves professional communication and saves valuable time for students and professionals alike.

The Problem

Professional writing is time-consuming, skill-dependent, and prone to errors — creating barriers for millions of users.

Our Solution

An AI web application powered by Llama 3.1 and Groq that generates polished, personalised documents instantly.

The Impact

Faster writing, professional quality, and personalised content — accessible to anyone, anywhere, at any time.

"Empowering everyone to communicate professionally".

Thank You

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